LLM-Assisted Risk-Adaptive Neurosymbolic Intent Planning for Optical Networks

A Pre-Deployment Decision Mechanism with Joint Semantic and QoT Assessment

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September 2026

OUTLINE

Motivation: The Evolution from Imperative SDON to Declarative IBON

⚠️ Current Paradigm: Imperative SDON
Procedural "HOW" Execution • Open-Loop Control

- Procedural Scripts
- Static Margins
- Open-Loop Control

PARADIGM SHIFT
"HOW" ➔ "WHAT"

⚡ Target Vision: Declarative IBON
Autonomous "WHAT" Abstraction • Closed-Loop Assurance

- ✓ High-Level Intents
- ✓ Dynamic Physics
- ✓ Zero-Touch Assurance

⚠️ BUT: Standard LLMs alone cannot simply drive an IBON controller

Illustrative Failure: Why Standard LLMs Break Optical Backbones

⚠️ 1. Token Budget Saturation	Telemetry dumps trigger attention degradation, dropping critical route exclusions
🚫 2. Hallucinated Physics	Probabilistic predictors lack wave propagation engines, violating non-linear GSNR margins
🔄 3. Semantic Drift	Unconstrained multi-turn conversational loops mutate or drop initial boundary constraints
⏱️ 4. Reactive Deployment Latency	Trial-and-error configuration risks live outages and introduces high control-plane recovery latency
👥 5. Suboptimal HITL Friction	Binary all-or-nothing review causes operator fatigue or outages; models fail to fail-early

Empirical Risk: Unconstrained LLMs might allow unfeasible deployments, semantic drift loops, and critical control-plane latency

Problem Statement: Given, Decide, Objective & Constraints

📥 [1] Given (System Inputs)

- Unstructured operator intent ($\mathcal{I}_{NL}$)
- Active optical topology graph $G(V, E)$
- Physical parameters (e.g. span length L)
- Feasibility threshold (e.g. $GSNR$)

⚙️ [2] Decide (Variables & Actions)

- Formal symbolic specification ($\mathcal{S}_{PDDL}$ compiled from intent)
- Optimal physical lightpath route ($\pi^* \in \mathcal{K}_{path}$ from candidate paths)
- Pre-deployment control action ($a \in \{approve, clarify, replan\}$)
- Provisioning routing configuration (c^* dispatched to controller)

🎯 [3] Objective (Optimization)

- Minimize human interruptions ($min\ N_{hitl}$)
- Minimize prompt tokens ($min\ T_{tokens}$)
- Minimize operational friction ($min\ \alpha \cdot N_{hitl} + \beta \cdot T_{tokens}$)
- Guarantee pre-deployment safety ($\mathcal{D}(U_{sem}, QoT_{valid}) = approve$)

🔒 [4] Constraints: Resource Limits vs. Physical & Semantic Boundaries

Resource Constraints (System & Solver Limits):

- Token context limits ($T_{prompt} \leq T_{max}$)
- Computational inference latency ($t_{exec} \leq t_{max\_budget}$)
- Symbolic solver complexity $K \in [3, 5]$

Boundary Constraints (Physical & Semantic Feasibility):

- Zero semantic drift tolerance bound
- Deterministic optical QoT feasibility
- Pre-deployment physical validity state ($QoT_{valid} \in \{0, 1\}$)

Proposed Solution: RADG with Neurosymbolic Planning

🧠 Neural Subsystem (Semantic Domain)

- Translation of the intent
- Validation of the semantic similarity
- Orchestration

📐 Symbolic Subsystem (Optical Domain)

- Topology extraction
- Deterministic tools usage
- Physical feasibility validation

🎯 Core Thesis Contributions

1. Scoped GraphRAG for IBN in a Neurosymbolic system

- Strict separation: probabilistic semantic translation vs deterministic physics
- Subtopology scoping reduces prompt tokens, eliminating attention loss

2. Quantification of Semantic Uncertainty for IBN in a Neurosymbolic system

- Layer 1 syntax check ($v_{struct} \in \{0, 1\}$)
- Layer 2 semantic discrepancy (d_{sem})

3. Risk-Adaptive Decision Gate (RADG) for optimized HITL

- Piecewise decision $D(U_{sem}, QoT_{valid})$: clarify, replan, or auto-approve
- Guarantees physical safety ($UAR = 0$) with <5 ms calculation latency

End-to-End System Architecture & Pipeline Flow

Phase 1

Optical RAG

Enrich $\mathcal{I}_{NL}$ with ITU-T grid & transponders

Phase 2

PDDL Parser

Translate intent into formal PDDL $\mathcal{S}_{PDDL}$ AST

Phase 3

Semantic Gate

[RISK GATE]

- Evaluate CFG v_{struct}
- Reverse Prompting d_{sem}

Phase 4

Symbolic Solver

Scoped GraphRAG & Yen's $K-SP$ routing

Phase 5

QoT Physics

Deterministic GN-model $GSNR$ calculation

Phase 6

Feasibility Gate

[RISK GATE]

Piecewise risk gate $D(U_{sem}, QoT)$

Phase 7

Plan Synthesizer

Auditable report & deployment commands

⏸ Phase 3b: Clarify Loop (LangGraph interrupt())

If $U_{sem} > \tau_{sem}$: Pauses execution and prompts operator for intent disambiguation

↺ Phase 6: Suggest Replan Loop (Physics Unfeasible)

If $QoT_{valid} = 0$: Suggests operator to relax constraints (lower baud rate, alternate link)

Overcoming Token Saturation: Scoped Optical GraphRAG

🌐 Deterministic Subtopology Scoping

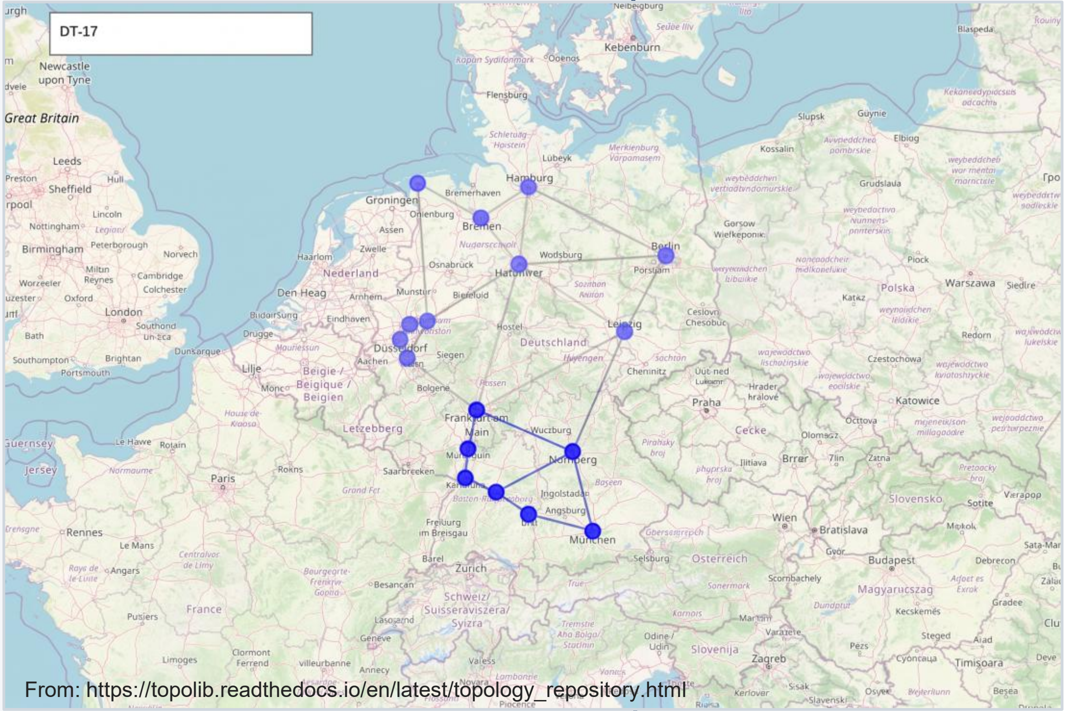
Raw JSON contains excessive telemetry: ROADM ports, EDFAs, fibers

Full topology dumps overwhelm LLM context windows

Mock GraphRAG extracts localized subtopology
$G_{sub} \subseteq G$

Mathematical Scoping Bound:

$$G_{sub} = (V_{sub}, E_{sub}) \subseteq G$$
$$T_{prompt}(G_{sub}) \ll T_{prompt}(G)$$



From: https://topolib.readthedocs.io/en/latest/topology_repository.html

e.g. Frankfurt ➔ Munich demand: Northern nodes pruned from LLM prompt

Overcoming Semantic Drift: Reverse Prompting & HITL

🧠 Two-Layer Semantic Uncertainty (U_{sem})

Structural
CFG regex AST checks
($v_{struct} \in \{0,1\}$)

Semantic
Reverse Prompting
reconstructs the intent

Independent LLM judge measures semantic discrepancy $d_{sem} \in [0,1]$

Uncertainty Formulation:

$$U_{sem} = \begin{cases} 1 & \text{if } v_{struct} = 0 \\ d_{sem} & \text{if } v_{struct} = 1 \end{cases}$$

⏸️ Fail-Fast HITL Clarification Loop

Fail-Fast Guarantee: Zero computational waste on physics simulation when intent is ambiguous

Pre-Deployment Risk Gate: The RADG Decision Function

Piecewise Decision Formulation:

$$D(U_{sem}, QoT_{valid}) = \begin{cases} clarify & \text{if } U_{sem} > \tau_{sem} \\ replan & \text{if } U_{sem} \leq \tau_{sem} \wedge QoT_{valid} = 0 \\ approve & \text{if } U_{sem} \leq \tau_{sem} \wedge QoT_{valid} = 1 \end{cases}$$

❓ Clarify Intent

$U_{sem} > \tau_{sem}$

- Trigger: Semantic uncertainty exceeds threshold
- Action: Pause pipeline via interrupt()
- Prompts operator to provide missing data
- Prevents unverified intent from reaching physics

↺ Suggest Replan

$U_{sem} \leq \tau_{sem} \wedge QoT_{valid} = 0$

- Trigger: Valid semantics, but physics infeasible
- Action: Physics failed; notifies operator
- Suggests relaxing constraints (e.g. lower baud rate)
- Loops back to PDDL parsing with feedback

✓ Auto-Approve

$U_{sem} \leq \tau_{sem} \wedge QoT_{valid} = 1$

- Trigger: Clear semantics and feasible $GSNR$
- Action: Autonomous zero-touch provisioning
- Generates auditable planning report
- Zero unverified configurations reach controller

Deterministic Physical Layer: GN-Model QoT Validation

🔬 Analytical Gaussian Noise (GN) Engine

- Pure Python implementation: zero LLM arithmetic

Accumulated ASE Noise Power per Span:

$$P_{ASE} = (G - 1) \cdot h \cdot \nu \cdot F \cdot B_{ref}$$

Nonlinear Interference (NLI) Noise Power:

$$P_{NLI} \approx \eta \cdot {P_{ch}}^{3}$$

- Accounts for fiber Kerr nonlinearity, dispersion, and EDFA noise

📐 Optical Feasibility & GSNR Criteria

Generalized Signal-to-Noise Ratio (GSNR):

$$GSNR = \frac{P_{ch}}{P_{ASE} + P_{NLI}} \geq GSNR_{th}$$

Receiver Power Sensitivity Budget:

$$P_{rx} = P_{launch} - A_{total} + G_{total} \geq P_{rx,min}$$

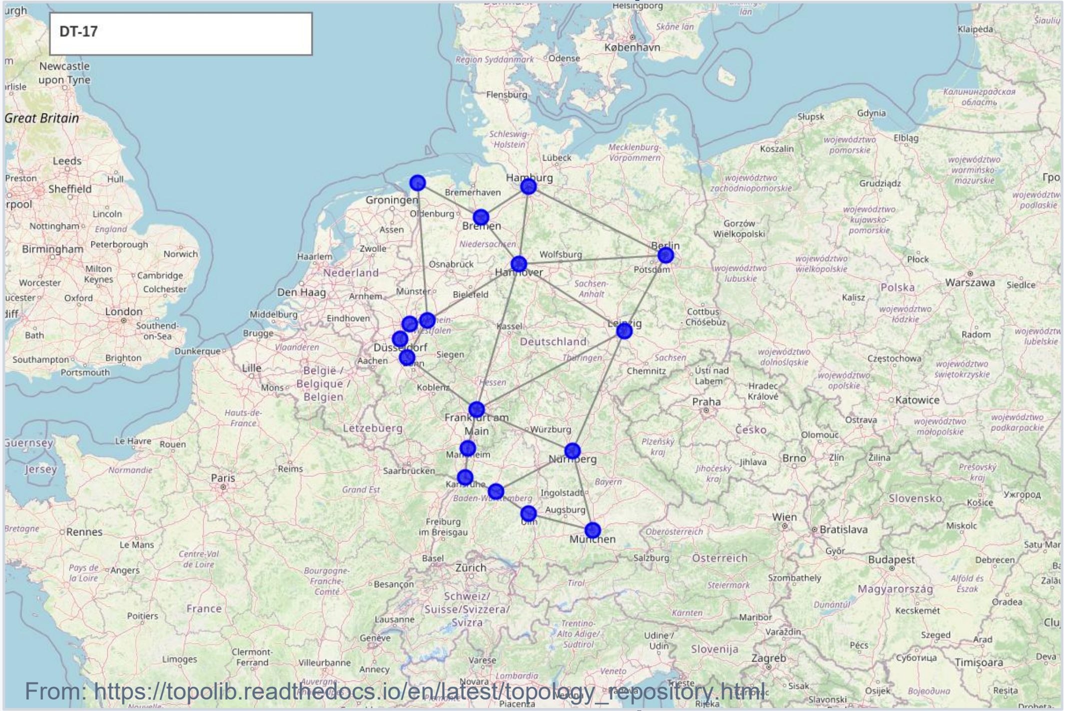
- Deterministic feasibility: $GSNR \geq GSNR_{th} \wedge P_{rx} \geq P_{rx,min}$

⚡ **Execution Speed:** $T_{phys} < 5\ ms$ **per candidate route (100% deterministic reproducibility)**

Experimental Setup & Testbed Environment

🌐 17-Node German Core Network Benchmark

- Realistic telecom topology: $|V| = 17, |E| = 26$ bidirectional fiber links
- Standard Single-Mode Fiber (SMF-28): $\alpha = 0.2\, dB/km$
- Dispersion parameter $D = 16.7\, ps/(nm \cdot km)$, $\gamma = 1.2\, W^{-1} km^{-1}$
- Amplified spans: dual-stage EDFAs with noise figure $NF = 5.5\, dB$
- Dynamic span lengths ranging from $L \in [45, 350]\, km$
- Orchestrator: LangGraph StateGraph with memory persistence
- Telemetry: RESTConf and Mock SDON testbed client adapters



From: https://topolib.readthedocs.io/en/latest/topology_repository.html

Evaluation Framework & Benchmark Scenarios

⚖️ Architectural Baselines

Baseline A
LLM-Only

vs.

Baseline B
Always-HITL

Proposed Neurosymbolic RADG
Decoupled translation + sequential pre-deployment risk gates

🧪 100 Test Demands (4 Risk Classes)

Class & Size	Intent Characteristics	RADG Action
Class I: Nominal [40]	Feasible path, unambiguous	**Auto-Approve**
Class II: Ambiguous [20]	Under-specified (U_sem > τ)	**Clarify Intent**
Class III: Infeasible [25]	Violates GSNR threshold	**Suggest Replan**
Class IV: Adversarial [15]	Hallucinated nodes (v_struct=0)	**Reject Intent**

🧠 1. Semantic Translation Accuracy (Neural Domain)

- Test Focus: Validates NL ➔ PDDL translation without constraint loss or hallucinations
- Key Metrics: Constraint Retention Rate (CRR = 100%) • CFG AST Pass Rate ($v_{struct} = 1$)

📐 2. Physical Feasibility (Optical Layer Integrity)

- Test Focus: Validates optical reach and non-linear impairments before controller push
- Key Metrics: Unsafe Approval Rate (UAR = 0% hard invariant) • QoT Feasibility (100%)

⚡ 3. Orchestration & Resource Efficiency (System Limits)

- Test Focus: Quantifies prompt token savings from GraphRAG and operator fatigue reduction
- Key Metrics: > 75% Token Reduction ($G_{sub} \subseteq G$) • > 70% HITL Cut • Sub-second compute

🔒 4. RADG Decision Robustness (Gate Reliability)

- Test Focus: Stress-tests piecewise decision logic (U_{sem}, QoT_{valid}) across boundary conditions
- Key Metrics: Gate Decision Accuracy (> 98%) • Zero False Positives (FPR = 0%)

Key Findings & Pre-Deployment Guarantees

100%

Pre-Deployment Safety Guarantee

$UAR = 100\%$: Zero unfeasible lightpaths reach controller (vs 34% in baseline)

> 70%

Reduction in Operator Fatigue

Autonomous approval when $U_{sem} \le \tau_{sem}$ (operator engaged only on high risk)

< 5 ms

Deterministic Physics Latency

GN model executes in $T_{phys} < 5\ ms$ ($T_{solver} < 10\ ms$, sub-second total)

Empirical Benchmark: GSNR Distribution & Blocking Rate vs Baselines

(Target: 16:9 Matplotlib / Chapter 4 Benchmark Plot — Replace in PowerPoint)

Conclusions & Main Takeaways

🧠 Neurosymbolic Decoupling is Essential

- Probabilistic LLMs must never calculate physical impairments or route lightpaths
- Natural language reasoning $\mathcal{I}_{NL} \rightarrow \mathcal{S}_{PDDL}$ bridges to formal planning
- Guarantees formal soundness without constraining operator expressiveness

🔒 Sequential Risk Gates Prevent Failure

- Early semantic evaluation ($U_{sem} \leq \tau_{sem}$) eliminates drift before physics computation
- Deterministic GN-model verification ($QoT_{valid} = 1$) ensures 100% physical feasibility
- Zero unverified configurations ever reach the optical controller

⚡ Selective HITL Optimizes Efficiency

- Engaging operators proportionally to risk eliminates fatigue while maintaining safety
- Over 70% reduction in human intervention compared to always-on review
- Sub-second planning latency enables rapid dynamic lightpath turn-up

⚙️ Production-Ready Engineering

- Fully implemented LangGraph state machine with memory checkpoints
- Benchmarked on realistic $|V| = 17, |E| = 26$ German topology and RESTConf testbed
- Establishes a reproducible foundation for autonomous optical networking

Future Outlook & Acknowledgments

🖥️ Joint Compute and Optical Scheduling

Extending PDDL domain to co-schedule distributed GPU cluster workloads alongside optical lightpaths

🌈 Multi-Band & Dynamic Optical Channels

Expanding analytical GN-models to incorporate C+L multi-band SRS inter-channel Raman crosstalk

🔄 Autonomous Online Self-Healing

Ingesting real-time testbed telemetry for closed-loop intent re-planning upon physical fiber degradation

ACKNOWLEDGMENTS

Sincere gratitude to Academic Advisor Prof. Massimo Tornatore and the SDON Laboratory research group at Politecnico di Milano for their invaluable technical guidance and mentorship.

Thank You for Your Attention!

Questions & Discussion